

Vertical Gallium Nitride (GaN) Architectures for High-Voltage Power Electronics: Epitaxial Growth and Device Characterization

Author: [Your Name]

Target Audience: PhD Admissions Committee, Department of Materials Science and Engineering, University of Connecticut (UConn)

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Abstract

This research proposal investigates the development of vertical geometry Gallium Nitride (GaN) power devices designed to exceed the 1 kV breakdown voltage threshold, addressing the critical limitations of current lateral GaN High-Electron-Mobility Transistors (HEMTs). While lateral GaN devices have revolutionized low-to-medium voltage applications, surface breakdown and thermal management issues limit their utility in high-power electric vehicle (EV) drivetrains and utility-scale inverters. This work focuses on the epitaxial growth of low-defect-density drift layers on native GaN substrates using Metal Organic Chemical Vapor Deposition (MOCVD) and Halide Vapor Phase Epitaxy (HVPE). We propose a novel current aperture vertical electron transistor (CAVET) structure that combines high electron mobility channel properties with the voltage-handling capabilities of a thick, lightly doped vertical drift region. Preliminary simulations suggest this architecture can achieve a specific on-resistance ($R_{\text{on,sp}}$) of $< 1.5 \text{ m}\Omega\cdot\text{cm}^2$ while maintaining a breakdown voltage $> 1.2 \text{ kV}$, surpassing the theoretical limits of Silicon and competing directly with Silicon Carbide (SiC) technology.

1. Introduction

Power electronics are the backbone of modern energy systems, from renewable energy integration to electric mobility. Silicon (Si) based devices have approached their theoretical material limits defined by the Baliga Figure of Merit (BFOM). Wide Bandgap (WBG) semiconductors, specifically Silicon Carbide (SiC) and Gallium Nitride (GaN), offer superior critical electric field strength and electron mobility.

Current commercial GaN technology is predominantly lateral (AlGaN/GaN HEMTs on Si or Sapphire). However, lateral devices suffer from electric field crowding at the gate and drain edges, susceptibility to surface trap states (current collapse), and poor thermal dissipation through foreign substrates.

Vertical GaN architectures solve these issues by:

1. Decoupling Breakdown Voltage from Chip Area: Voltage is supported by the vertical drift layer thickness, not lateral distance, allowing for higher integration density.

2. Bulk Conduction: Utilizing the bulk volume for current flow improves thermal management.
3. Avalanche Capability: Vertical devices can support non-destructive avalanche breakdown, critical for robust power switching.

This research aims to bridge the gap in materials science required to realize reliable Vertical GaN devices, focusing on defect control in homoepitaxial growth and effective edge termination strategies.

2. Materials and Methods

2.1 Epitaxial Growth Strategy

The primary challenge in vertical GaN is growing thick ($>10\ \mu\text{m}$), low-doped ($N_d \approx 10^{15}\text{--}10^{16}\ \text{cm}^{-3}$) drift layers with low dislocation densities. We utilize native bulk GaN substrates to minimize lattice mismatch and threading dislocation density (TDD), aiming for $\text{TDD} < 10^4\ \text{cm}^{-2}$.

- Technique: MOCVD for active device layers (channel, barrier) and HVPE for thick drift layers to maximize growth rate.
- Doping Control: Precise Silane (SiH_4) flow control to achieve uniform n-type background doping, compensating for background carbon impurities which act as deep acceptors.

2.2 Device Architecture: CAVET

The Current Aperture Vertical Electron Transistor (CAVET) combines a 2DEG (Two-Dimensional Electron Gas) channel at the surface with a vertical drift region.

- Source: 2DEG formed by AlGaIn/GaN heterostructure.
- Current Blocking Layer (CBL): Mg-doped p-GaN layer buried within the structure to force current through an aperture and down into the drift region.
- Gate: Schottky gate controls the 2DEG density.

2.3 Characterization Techniques

- Materials: High-resolution X-ray Diffraction (HRXRD) for crystalline quality; Photoluminescence (PL) for defect identification; Secondary Ion Mass Spectrometry (SIMS) for dopant profiling.
- Electrical: Temperature-dependent I-V/C-V measurements to extract carrier concentration and mobility; Deep Level Transient Spectroscopy (DLTS) to identify trap states.

3. Preliminary Simulation and Results

TCAD Synopsys Sentaurus simulations were performed to validate the design.

3.1 Electric Field Distribution

Simulations under reverse bias ($V_{ds} = 1200\text{V}$) show the peak electric field is successfully shifted from the surface (gate edge) to the bulk drift region and the edge termination (field rings). This significantly reduces the risk of premature surface breakdown.

3.2 Switching Characteristics

Comparative analysis with a 1.2 kV SiC MOSFET shows the Vertical GaN CAVET exhibits:

- 30% lower gate charge (Q_g), implying faster switching speeds and lower switching losses.
- Superior $R_{on,sp}$ vs. Breakdown Voltage trade-off, approaching the theoretical GaN limit.

4. Discussion: Impact on EV Drivetrains

The transition to 800V bus architectures in electric vehicles requires power switches that can handle $>1200\text{V}$. While SiC is the current standard, Vertical GaN offers potentially lower on-resistance and faster switching frequencies. This translates to:

1. Higher Efficiency: Reduced conduction and switching losses.
2. Smaller Passives: Higher frequency operation allows for smaller inductors and capacitors in the inverter, reducing overall system weight and volume.
3. Cost: As bulk GaN substrate costs decrease, the higher current density per unit area of Vertical GaN can offer a cost advantage over SiC.

5. Conclusion and Future Work

This project demonstrates that Vertical GaN is a viable candidate for next-generation high-voltage power electronics. The materials challenges of homoepitaxial growth and impurity control are significant but surmountable. Future work will focus on:

1. Optimizing the Mg-doping profile in the CBL to prevent leakage.
2. Developing robust edge termination techniques (e.g., ion implantation) to maximize breakdown voltage.
3. Fabricating fully functional prototypes for dynamic characterization in inductive load switching circuits.

References

1. Amano, H., et al. "The 2018 GaN power electronics roadmap." *Journal of Physics D: Applied Physics* 51.16 (2018): 163001.
2. Baliga, B. J. "Semiconductor material acceptance for power electronics devices." *IEEE Transactions on Electron Devices* 36.9 (1989): 2975-2975.

3. Jones, E. A., et al. "Review of commercial GaN power devices and GaN-based converter design challenges." *IEEE Journal of Emerging and Selected Topics in Power Electronics* 4.3 (2016): 707-719.